Elderly Home Activities

Goals:

Elderly folks are often under-represented in educational efforts revolving around STEM, as many focus on youth learning. This is because of a number of reasons, but among them is the emphasis on scientific learning as a vessel for productivity rather than something inherently fulfilling. This disparity in educational outreach has led to discrepancies in vaccine hesitancy, patient nonadherence, and trust in the scientific community. In hopes of helping amend these mistakes, we want to help seniors hold a better understanding of synthetic biology. At the same time, we want to encourage a productive dialogue around fears in synthetic biology. This event focuses on teaching the components of human cells, including organelles, their functions, and impacts on medicine and technology.

The Ask-Teach-Talk Model:



In order to accomplish all of our goals and get feedback to iterate on, we adopted an ask-teach-talk model into our plans. First we ask participants to fill out an optional survey attached at the end of this document. This survey asks for opinions around science, health, and technology. We hope that by incorporating direct feedback, that we can improve our approach and measure our impact. We then begin the activity by discussing our topic and engaging in an activity. By first giving an overview of the topic, we aim to give participants the tools they need to better understand synthetic biology and better communicate both enthusiasm and anxiety. Then going into the discussion we can engage with both. Our discussion hopes to ease anxieties around synthetic biology, while also bridging the gap between academia and the general public. Finally, at the end we give a similar survey to see the impacts of our approach, and give time to reflect for participants.

Our Activity:

After beginning with our survey, we will begin with the presentation attached to this document. We will then hand out a variety of bingo boards, containing pictures that match with words we will mention throughout the presentation. We will encourage the participants to fill out the bingo board when they hear the term that corresponds to the pictures on their sheets. The first person to get a bingo will receive a reward. We first discuss the central dogma of biology, how DNA stores information, RNA communicates that information, and proteins act as molecular machinery. Moving onto examples of this in the real world, we introduce antibiotics. Discussing how beta lactams inhibit proteins involved in the synthesis of the cell membrane, paired with a diagram using the similar “molecular machinery” concept. We can also give mRNA vaccines as an example of central dogma in action. We will explain how mRNA vaccines use mRNA to

express harmless SPIKE proteins on human cells, until the RNA is degraded in order to train our immune system. Finally, blood thinners that act as folate antagonists fit into the same spot as regular folate, but without function, preventing proper blood clotting.

We then turn our discussion to cell signaling. Using a new motif of discussing signals as communication between cells and organs. This can help show that more complicated pathways can be inefficient. Following this, we can talk about how cells can become over/undersensitized. If too many balls are in one bucket already, it cannot take more balls, and we can show that visually. An example we want to give is GLP-1 agonists, as it commandeers these signaling pathways to give a false sense of fullness and an insulin spike. We can also mention how even 1 signaling molecule can cause a lot of different effects, which is why GLP-1 agonists can cause nausea and stomach issues. We can tie this back to the analogy we made earlier, since many balls spilled many effects occurred from what we might assume is a straight forward pathway. We can then discuss anti-histamines, which blocks the binding of histamine, a molecule that signals an immune response. At the end, we can combine both of our lessons to discuss how Advil works. Advil stops proteins that produce signaling molecules that cause fever and pain. So instead of blocking the “listener” cell, it stops the “speaker” cell from talking.

Discussion:

In order to facilitate a productive discussion, we will have prompting questions in the slideshow. These questions are meant to spark productive discussion using the vocabulary we helped develop here, but to communicate fears around new technologies. We hope that by establishing ourselves as open to communication, we can properly hear experiences and bridge the gap between scientists and elderly folk.

Survey Form

What does the phrase synthetic biology mean to you?

Rank the following statements from 1-5, 1 being strongly disagree and 5 being strongly agree.

Statement	Strongley Disagree				Strongley Agree
Scientists are actively involved members of my community	1	2	3	4	5
I believe scientists have the public's best interest in mind	1	2	3	4	5
I am skeptical of science	1	2	3	4	5
I understand current advancements in science	1	2	3	4	5
Scientists listen to my concerns	1	2	3	4	5
This activity was exciting and interesting	1	2	3	4	5
I left this activity with a better understanding of science than I came in with	1	2	3	4	5
I felt the discussion eased my concerns with science	1	2	3	4	5

Write any further comments below the following questions:

BINGO Board Contents

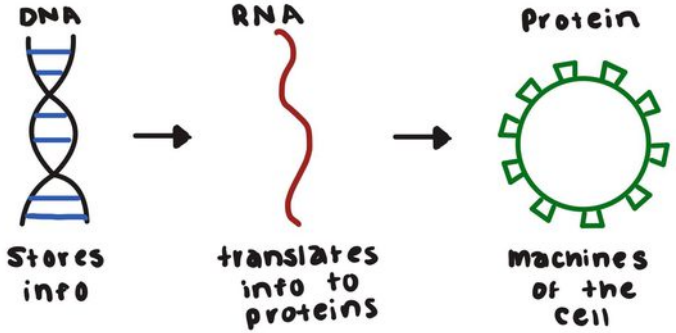
Word bank:
Cell-Singaling
DNA
RNA
Protein
Enzymes
Molecular machinery
Virus
mRNA
Allergies
Ozempic
Penicillin
Folate
Insulin
Immune system
Fever
Pain
Cell membrane
Brain
Liver
blueprint
Free space (synthetic biology)

New Synthetic Biology

Washington iGEM



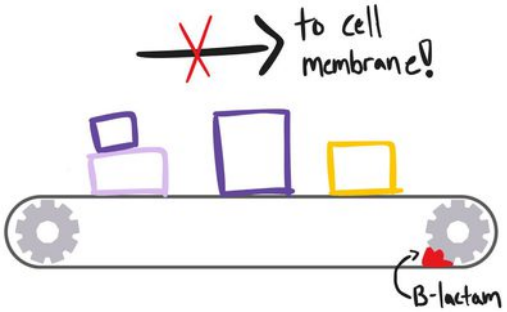
Genetics



- DNA acts as the blueprint
- RNA acts as the communicator
- Proteins acts as machines
- Not all parts of the blueprint are read\
- Genes can be turned on/off
- Some proteins called enzymes process other chemicals

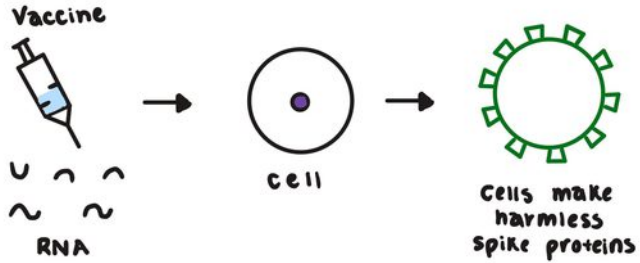
Antibiotics

- Beta lactams Interfere with cell membrane synthesis
- Includes:
  - Penicillin
  - Cephalosporin
  - Monobactam
  - Carbapenem
  - Clavam



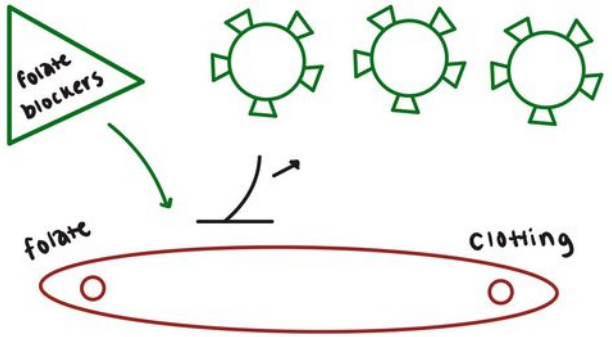
mRNA Vaccines

- RNA is injected into cells
- RNA codes for harmless spike proteins
- Some cells express these until RNA runs out
- Immune system learns



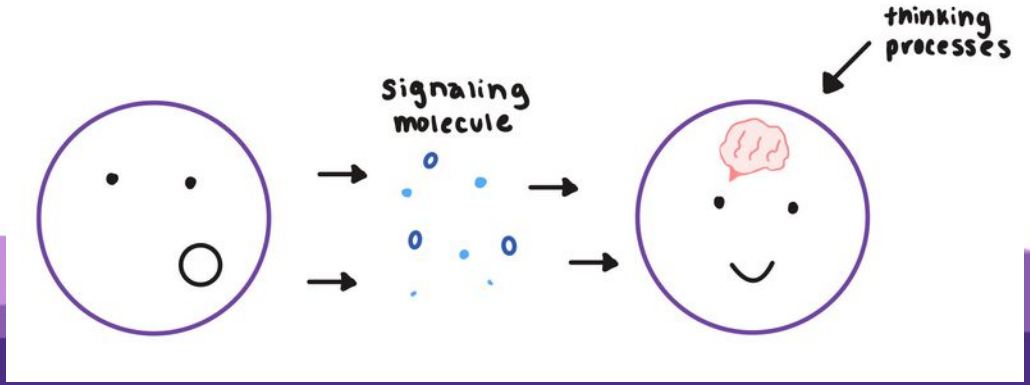
Blood Thinners

- Some blood thinners like methotrexate and trimethoprim block folate
- Folate is an early molecule that eventually can cause clotting
- Blocking folate prevents clotting



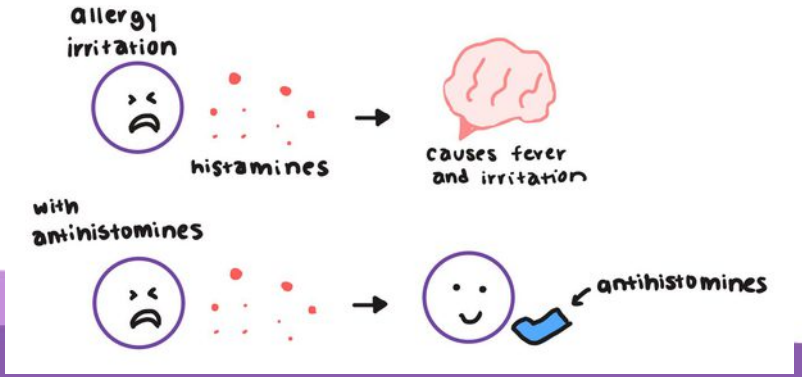
Cell Signaling

- Cells communicate with each other using chemicals
- One cell “speaks”(releases) a molecule
- Another cell “hears”(binds) that same molecule
    - This is usually paired with a signal cascade (cell thinking)
- Requires that molecule to be released and bind



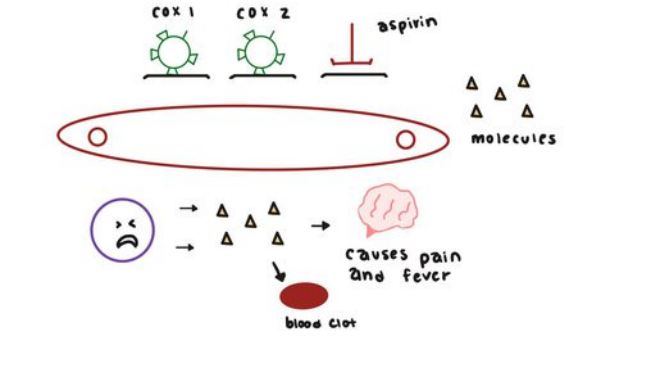
Allergies/Histamines

- Pollen irritation causes cells to release histamines
- Histamines cause fever/irritation
- Antihistamines blind to histamine receptors
- Histamines cannot bind



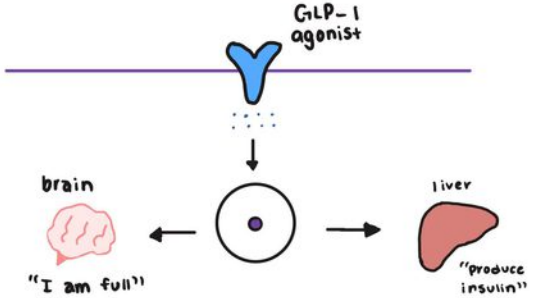
Aspirin

- All signaling molecules are formed from other molecules (usually from food)
- Aspirin inhibits COX 1 and COX 2 which produce signaling molecules
- Some molecules cause pain and fever
- Some molecules are needed for blood clotting



GLP-1s

- GLP-1 agonists bind to GLP-1 receptors in cells
- These cells communicate with the brain and liver
- Cause the effects of being full without food



Discussion!

When you hear about science in the news, what do you think?

What excites and what scares you about science?

Do you feel like your voice is heard in science?